Syllabus : Data Science for Social Impact

Summer 2025

Instructors: **DSI Preceptors** | Office: Ryerson 256

Class Schedule: Monday-Friday 09:00 PM-5:00 PM | Location: Ryerson Phys Lab 251

---

1 Curriculum Description

Welcome to DSSI 2025!
In the coursework dedicated for this year's fellowship, we will train students in a background of Machine Learning and some key areas of Natural Language Processing with a view to handling a variety of language tasks that will be explored during the research weeks of the program.

Natural Language Processing (NLP) is an area of computation that comprises of statistical models used to analyze and synthesize human languages. With the recent abundance of text data available across a variety of platforms, NLP has seen increased importance to automate a wide range of language tasks with applications across several fields like social science, business, medicine, psychology, and natural sciences.

The DSSI curriculum will begin with an overview of popular Machine Learning models that can be used to solve classification problems. We then proceed to cover the fundamentals of modern natural language processing, with a special emphasis on state-of-the-art models based on deep learning. These include concepts like language models, word embeddings, recurrent neural networks (Simple RNNs, LSTMs), and attention-based models such as the transformer and BERT.

We will use Python and Python-based libraries such as PyTorch, NLTK, and SpaCy to implement algorithms for a variety of text processing problems such as text classification, token classification, and text generation.

2 Curriculum Overview

The DSSI instructional period includes a two-week long course that will incorporate a few different teaching and assessment modalities. The course is supported by an experienced teaching staff of several instructors, and two teaching assistants. We will host a series of lectures to help students gain a thorough understanding of the numerous problems related to the processing and analysis of textual data. These lectures will often be accompanied by active learning through additional lab sessions. Students will also be required to demonstrate their proficiency in the topics discussed in class by completing additional programming assignments distributed as homework.

Attendance and participation are expected during all class meetings. We also expect respectful behavior from students towards their peers and all teaching staff.

Course Topics: We will cover a host of topics related to Machine Learning and Natural Language Processing for text analysis. These topics include but are not limited to:

- Python
  - Pandas and Python Primer
  - Python Dictionaries
  - Visualizations with Matplotlib
- Data Science
  - Probability and Empirical Distributions
  - Basic Statistical Inference
  - Classification: KNN and Logistic Regression
  - Training, Testing, Evaluation Metrics, Overfitting
  - Ensemble methods + Crossvalidation and Hyperparameter Tuning
  - Basic Feed Forward Neural Network
- Linear Algebra
  - Matrix Operations
  - Gradient Descent
  - Introduction to Tensors with PyTorch
- Natural Language Processing
  - N-grams and Language Modeling
  - Vector Semantics and Embeddings
  - Neural Language Modeling
  - RNN and LSTMs primer
  - Attention with Seq-2-Seq models
  - Transformers and LLM and Masked Language Models
  - Sequence Labeling for Parts of Speech and Named Entities
  - Neural Token Classification

A detailed course calendar laying out a tentative lecture schedule will be posted on Canvas and updated as needed.

Bibliography: We will not be referring to any specific textbook. We will publish lecture slides and any additional reading materials that will be assigned during the instructional period on Canvas.

Software: You will be coding in Python to solve the problems listed in the programming assignments as well as to implement the projects for this course. Hence it is essential that you have regular access to a machine with Python 3.7+ installed in it and a suitable software to create and edit Python scripts and notebooks. Typically we suggest installing Jupyter Lab, VSCode, or some other IDE. We will also be using Google Colab to work on the NLP related labs and assignments. Lastly, in this class you will be using GitHub for project management and version controlling. Some necessary links to help install the tools to be used for this course:

- Git: `https://docs.github.com/en/get-started/getting-started-with-git/set-up-git`

- VSCode: `https://code.visualstudio.com/docs/setup/setup-overview`
- Jupyter Notebooks in VS Code:
  `https://code.visualstudio.com/docs/datascience/jupyter-notebooks`
- Jupyter: `https://jupyter.org/install`

Announcements: All announcements related to this course will be posted on Canvas. *We will assume that any announcement made on Canvas is known to everyone in class within **half a business day** of being posted.* As such, it is important to check both Ed and Canvas regularly. You are responsible to be cognizant of all information disclosed in person during lectures or published online.

Office Hours and Meetings: Office hours with TAs will be held following classes during each day of instruction. This should be used to resolve inquiries on topics covered in lectures and programming assignments. Office hour times and locations of the TAs will be posted on Canvas.

3 Curriculum Policies

Class Participation and Attendance: You are **required** to attend classes and participate in class discussions. Missing a class during the instructional period is heavily discouraged, unless explicit permission has been sought from the DSSI instructors.

Note that if you are facing extenuating circumstances like illness or other emergencies, that keep you from attending classes or important session, please let us know as soon as possible so we can draw a plan to help you catch up with missed lessons.

Assessment Modules: To aid better understanding of curriculum materials and better prepare students for the research they will be embarking upon, the instructional weeks will distribute a number of **Programming Assignments**. You are required to work on them individually and they involve coding in Python using a Jupyter Notebook or Google Colab, as well as performing analysis using techniques covered in lectures. You are still allowed to discuss ideas and concepts covered in these assignments with your peers but the work that you submit should be strictly yours. You need to complete all the assignments before moving on to research work. You will be submitting these assignments via Gradescope and more information regarding that will be released in due time.

Use of AI Tools: Students are only allowed to use Generative AIs, such as ChatGPT or Google Bard, on assignments in this course when permission has been given by the instructor in advance. Students must submit an explicit request with an explanation of how they will use Generative AI for their assignments. Students are not permitted to use these tools until permission is granted in writing. The instructor may encourage and give permission to students to use AI tools during class activities and in other contexts when it is considered in support of the course learning goals. However, such consents will be granted entirely as per the instructor's judgement.
Unless given permission to use these tools, each student is expected to complete course assignments without assistance from AI tools. If you are unclear on whether something is an AI tool, please check with your instructor. Unauthorized use of AI tools for any purpose in this course will violate the DSSI's integrity policy.